SAEINDIA™

AEROSPACE FORUM

AeroTHON 2026

Track 1

ROTORCRAFT SYSTEMS CHALLENGE (UAS)

DESIGN, BUILD AND FLY CONTEST

Theme: Aerial Surveillance and Rapid Delivery

Rule Book

Revision 1
March 2026

REVISION HISTORY

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FOREWORD

Welcome to SAEINDIA AEROTHON – ROTORCRAFT SYSTEMS CHALLENGE (UAS) – DESIGN, BUILD AND FLY CONTEST 2026. The system requirements are developed to align with real-world Uncrewed Aerial Vehicle requirements to provide industrial exposure to Students.

The contest is planned in two phases:

1. Phase 1: Design Report & Presentation
2. Phase 2: Flying Competition

The student teams must submit a design report of their UAS in Phase 1, adhering to the contest design rules and guidelines, and give a presentation to the jury. The top performing teams from Phase 1 will qualify for Phase 2 of the contest in which the qualified teams are required to build an Uncrewed Aircraft System and successfully complete the missions as described in the rulebook during the flying competition. Winners will be awarded as per the announcement made by the Organising Committee. Please refer to AeroTHON 2026 webpage on SAEINDIA site.

Universities/Institutions can nominate any number of teams as long as they meet the team formation requirements listed in this document.

Lastly, contesting teams are requested to pay special attention to the bold and italicized fonts throughout this document. These are important updates and clarifications on a variety of aspects pertaining to the design. Please read these rules carefully. Watch out for official announcements and updates concerning this contest and rule interpretations on SAEINDIA website.

Best of luck to you all!!

SAEINDIA Aerospace Forum
SAEINDIA
AEROTHON 2026 Organising Committee

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1.0. CONTEST DETAILS

1.1 OVERVIEW

AeroTHON 2026 continues to be a crucial initiative for engineering students in India as the country accelerates its push to become a global leader in Uncrewed Aircraft Systems (UAS) and drone technologies. This vision aligns with the Government of India's Atmanirbhar Bharat Abhiyan, which emphasizes indigenous development and self-reliance in critical technologies, including drones and autonomous aerial systems.

Globally, the UAS industry is experiencing rapid expansion, driven by advancements in artificial intelligence, sensing technologies, autonomy, and connectivity. Drones are increasingly being deployed across a wide range of sectors such as agriculture, infrastructure inspection, logistics, disaster response, environmental monitoring, and defense operations. The growing demand for unmanned aerial systems is transforming industries by improving efficiency, safety, and operational capabilities.

India's drone ecosystem is also expanding rapidly, supported by progressive regulatory reforms, government initiatives, and increasing adoption across commercial, industrial, and public sectors. The country's drone market is expected to witness significant growth in the coming years, driven by technological innovation, expanding applications, and supportive policies that encourage indigenous development and manufacturing of drone technologies.

In parallel, the Government of India has introduced several initiatives to promote domestic drone innovation and manufacturing, including the Production-Linked Incentive (PLI) scheme, "Drone Shakti" initiatives, and expanded research and development programs. These initiatives aim to strengthen the domestic drone ecosystem, encourage local technology development, and position India as a global hub for drone innovation.

In this context, the **SAEINDIA Aerospace Forum** is organizing the **SAEINDIA AeroTHON 2026 – Rotorcraft Systems Challenge**, a Design–Build–Fly competition for engineering students. The primary objective of this challenge is to develop students' capabilities in designing, building, and operating unmanned rotorcraft systems, preparing them to contribute effectively to the rapidly evolving drone industry.

This competition provides undergraduate and graduate engineering students with a **real-world engineering experience**, exposing them to the practical challenges faced by engineers in the aerospace and drone sectors. Participating teams will conduct trade studies, perform engineering analyses, and make design decisions to develop a rotorcraft system that satisfies the defined mission requirements and system specifications.

As part of the competition, teams will design, build, integrate, and demonstrate a flight-worthy UAS capable of successfully performing the prescribed mission tasks during the flight demonstration event. This process requires students to apply multidisciplinary engineering principles, including aerodynamics, propulsion systems, structures, avionics integration, control systems, and systems engineering.

In essence, AeroTHON 2026 – Rotorcraft Systems Challenge offers students a unique opportunity to experience the complete engineering lifecycle of a UAS, from conceptual design and system architecture to subsystem integration, testing, validation, and final flight demonstration.

Through initiatives such as AeroTHON, SAEINDIA aims to nurture the next generation of aerospace and drone engineers in India. As the nation progresses toward becoming a global hub for drone technology, there is a growing need for skilled professionals capable of designing, developing, and operating advanced unmanned systems. By providing students with a hands-on platform to innovate, experiment, and demonstrate real-world engineering solutions, AeroTHON plays a vital role in strengthening India's drone ecosystem and supporting the country's vision of technological leadership in autonomous aerial systems by 2030.

The importance of practical and interpersonal communication skills is often overlooked by engineers. It is important to note that, apart from technical knowledge, written and communication skills are vital in the engineering workplace. To help the students develop these skills, the contest has been divided into two phases:

1. Phase 1: Design Report & Presentation
2. Phase 2: Flying Competition

Phase 1: Design Report & Presentation

In the first phase, students will focus on designing a UAS that adheres to current technical standards and mission requirements. The trade-off decisions made here reflect the cutting-edge technology being developed globally. Students will also focus on producing a detailed design report and presentations, reflecting the growing importance of documentation and communication skills in the engineering workplace, whether presenting to stakeholders, clients, senior engineers or regulatory authorities.

Phase 2: Flying Competition

The second phase will require students to build and test a working UAS that meets the mission requirements. In a competitive environment where safety and performance are paramount, the students' designs will be put to the test. Drones capable of performing complex tasks like thermal imaging, real-time data processing, and autonomous disaster situation identification will be key to success.

1.2 OBJECTIVES

1. To inculcate innovation mindset among the student community in emerging technologies like Uncrewed aerial vehicles (UAS)
2. To incubate and nurture aeronautical design skills and engineering capabilities in young minds, preparing them to contribute to the vision of **Atmanirbhar Bharat** in critical aerospace technologies
3. To provide a platform for aerospace-passionate students to demonstrate their expertise in UAS design, development, and system integration.
4. To inspire and support the development of the next generation of innovators and entrepreneurs in the aerospace and drone ecosystem.

1.3 RULES AND ORGANIZER AUTHORITY

1.3.1 General Authority

SAEINDIA AEROTHON – ROTORCRAFT SYSTEMS CHALLENGE (UAS) – DESIGN, BUILD AND FLY CONTEST 2026 committee reserves the right to revise the schedule of any contest and/or interpret or modify the contest rules at any time and in any manner that is, in their sole judgment, required for the efficient operation of the event.

1.3.2 Rules Authority

SAEINDIA Aerospace Forum owns the responsibility and authority of the rules of SAEINDIA AEROTHON – ROTORCRAFT SYSTEMS CHALLENGE (UAS) – DESIGN, BUILD AND FLY CONTEST 2026 and it has been issued under the authority of the SAEINDIA. Official announcements from the SAEINDIA AEROTHON – ROTORCRAFT SYSTEMS CHALLENGE (UAS) – DESIGN, BUILD AND FLY CONTEST 2026 Organizing Committee shall be considered part of and have the same validity as these rules. Ambiguities or questions concerning the meaning or intent of these rules will be resolved by the officials, SAEINDIA AEROTHON – ROTORCRAFT SYSTEMS CHALLENGE (UAS) – DESIGN, BUILD AND FLY CONTEST 2026 Organizing Committee or SAEINDIA Staff.

1.3.3 Rules Validity

The SAEINDIA AEROTHON – ROTORCRAFT SYSTEMS CHALLENGE (UAS) – DESIGN, BUILD AND FLY CONTEST 2026 Rules posted on the SAEINDIA Website and dated for the calendar year of the contest are the rules in effect for the contest. Rule sets dated for other years are invalid.

1.3.4 Rules Compliance

By entering the SAEINDIA AEROTHON – ROTORCRAFT SYSTEMS CHALLENGE (UAS) – DESIGN, BUILD AND FLY CONTEST 2026, the team members, faculty advisors and other personnel of the participating university/institute has agreed to comply with and be bound by the rules, interpretations or procedures issued or announced by SAEINDIA AEROTHON – ROTORCRAFT SYSTEMS CHALLENGE (UAS) – DESIGN, BUILD AND FLY CONTEST 2026 Committee. All team members, faculty advisors and other university representatives are requested to cooperate and follow all instructions from the contest organizers, officials and the jury.

1.3.5 Understanding the Rules

The participating student teams are responsible for reading and understanding the rules in their entirety, their effect on the contest in which they are participating. The section and paragraph headings of these rules are provided to facilitate reading and will not affect the paragraph contents.

1.3.6 Consideration of "Participation" in the contest

Teams, team members as individuals, faculty advisors, and other representatives of a registered university who are listed as team members while registering their team are considered to be "participating" in the contest from the time they register for the event until the conclusion of the contest or earlier in the case of withdrawal.

1.3.7 Violations of Rule Intent

The violations of the intent of a rule will be considered a violation of the rule itself. Questions about the intent or meaning of a rule may be addressed to the SAEINDIA AEROTHON – ROTORCRAFT SYSTEMS CHALLENGE (UAS) – DESIGN, BUILD AND FLY CONTEST 2026 Committee or SAEINDIA Staff.

1.3.8 Conditions and Penalties

Organizers have the right to modify the points and/or penalties listed in the various event descriptions to better reflect the design of their events, or any special conditions unique to the contest.

1.3.9 Mode of Communication

The The official mode of communication for the event will be through the registered email address provided on the event website: https://saeindia.org/aerothon2026/. All teams are required to regularly monitor communications sent to their registered email address.

Any communication through WhatsApp groups or similar messaging platforms is intended only for convenience and for sending alerts. Such platforms will not be considered an official channel for important announcements or instructions. Teams are advised not to contact industry experts, judges, or event officials related to the competition through social media or any other unofficial communication channels. Any such attempt to reach out may be considered a violation of the competition rules, and appropriate action may be taken against the team.

1.3.10 Force Majeure

- The AeroTHON 2026 Organizing Committee and SAEINDIA shall not be held responsible for any failure or delay in fulfilling their obligations under this agreement if such failure or delay is caused by the occurrence of one or more Force Majeure Events. These events include, but are not limited to, acts of God, war, floods, earthquakes, strikes, lockouts, pandemics, epidemics, riots, civil commotion, or scarcity of essential utilities such as water, electricity, or other basic facilities.
- The organizing committee shall inform the participating teams about the occurrence and cessation of such events **within one week of the decision being made.**
- If conducting the event becomes infeasible due to an extended duration of Force Majeure conditions or any other unforeseen circumstances beyond the control of the organizers, **the event may be postponed, modified, or cancelled for the year.**

Force Majeure Events include, but are not limited to:

- Natural disasters such as earthquakes, floods, inundation, landslides, storms, tempests, hurricanes, cyclones, lightning, thunder, pandemics, epidemics, extreme atmospheric disturbances, or any other **act of God**.
- Strikes, labour disruptions, or industrial disturbances not arising due to the acts or omissions of the organizers; war, hostilities (declared or undeclared), invasion, acts of foreign enemies, terrorism, rebellion, riots, armed conflict, military actions, civil war, ionizing radiation, contamination by radioactivity from nuclear fuel or waste, radioactive or toxic explosions, volcanic eruptions, or any similar events beyond the reasonable control of the organizers.
- Acts of expropriation, compulsory acquisition or takeover by any government agency of the said venue where the event is to be held or any part thereof
- Any prohibitory or restraining order issued by a **court of law or competent government authority** preventing the conduct of the event.

1.4 ELIGIBILITY

1.4.1 Team member

Members of a Team must be undergraduate or postgraduate student, and every member of the team must be a member of SAE India.

1.4.2 Society membership

A university or institute can nominate as many teams as they wish by paying the requisite fee for each team. However, each team must work independently.

The registration fees indicated in the Section 1.7 are required to be paid within 15 days of registration for offline registration. For online registrations, the fee payment must be made at the time of registration.

1.5 OFFICIAL LANGUAGES

The official language of the SAEINDIA AEROTHON – ROTORCRAFT SYSTEMS CHALLENGE (UAS) – DESIGN, BUILD AND FLY CONTEST 2026 is English. Document submissions, presentations, and discussions in English are acceptable during all the phases of the contest.

1.6 CONTEST PHASES

1.6.1 Phase 1: Design Report & Presentation

- This phase invites innovative designs from the participant teams
- The innovative designs will be evaluated by industry and academic experts
- The top performing teams in Phase 1 will be shortlisted for Phase 2.

The AEROTHON organizing committee reserves the right to select the top-performing teams from Phase1 to advance to Phase 2, with the number of teams varying based on performance in Phase 1.

1.6.2 Phase 2: Flying Competition

- Students will build a **fully functional physical model** based on the design presented in **Phase 1**.
- A **flight test demonstration** will be conducted to evaluate the performance of the UAS.
- **Awards will be presented to the top performing teams** based on their overall performance in the competition.

IMPORTANT DATES

Kindly refer to the AeroTHON 2026 website for the latest updates on the dates and timelines (https://saeindia.org/aerothon2026/)

*SAEINDIA AEROTHON – ROTORCRAFT SYSTEMS CHALLENGE (UAS) – DESIGN, BUILD AND FLY CONTEST 2026 Organizing Committee reserves right to alter any of the dates. Watch out the website for the latest updates.

1.7 REGISTRATION AND FEES

A team can comprise a minimum of 5 students and a maximum of 10 students of multiple disciplines with one faculty advisor. Please note that all student participants must be SAEINDIA members to participate in events or contests by SAEINDIA. Faculty advisors are advised to become members of SAEINDIA, though it is not mandatory.

The **Registration fee** for AEROTHON is **Rs.20,000/- (Rupees Twenty Thousand only) per team excluding 18% GST**. To register for AEROTHON and registration guidelines visit: https://saeindia.org/aerothon2026/

Steps to become a SAEINDIA Member

If you are not a SAEINDIA member, go to www.saeindia.com and select the "Membership" link. Students need to select the "Student Membership" link and provide the details as indicated. Alternate link to sign up for SAEINDIA membership https://www.saeindia.org/become-a-member. Faculty members who wish to become SAEINDIA members should choose the "Professional Membership" link.

1.8 CANCELLATION OF CONTEST REGISTRATION

Registration fees are not refundable and non-transferable to other teams or to subsequent year's competition. Teams registering for SAEINDIA AEROTHON – ROTORCRAFT SYSTEMS CHALLENGE (UAS) – DESIGN, BUILD AND FLY CONTEST 2026 are required to submit a design report on the design of the UAS. Failure to submit the Design report on or within the specified date will constitute an automatic withdrawal of your team from the contest. Your team will be notified the next day of the due date about non-submission; your team's registration will be cancelled two days after this notification and no refund will be given.

1.9 EXPECTATIONS

1.9.1 Design with no professional's involvement

The UAS must be designed solely by SAEINDIA student members, without direct involvement from faculty members or external professionals in the design process. Students may refer to literature or publicly available information related to UAS or aircraft design and construction. They may also seek guidance from industry mentors or professors; however, such interactions must be limited to discussions of design alternatives, including their advantages and disadvantages.

External professionals, mentors, or faculty members shall not make design decisions, prepare drawings, contribute to the design report, or participate in the construction of the UAS.

The Faculty Advisor must review the team's work and sign the Statement of Compliance provided in Appendix A, confirming that the design and development work has been carried out solely by the student team members in accordance with the competition rules.

1.9.2 Original design

- Any UAS presented in the contest must be an original design, with the configuration conceived and developed by the student team members. Replication or direct copying of existing UAS designs is not permitted.
- **Photographic scaling, reverse engineering, or direct reproduction of any existing model UAS design is strictly prohibited.** Teams must demonstrate that their design is the result of their own conceptual development and engineering analysis.

1.9.3 Unique designs

Universities or institutions may register more than one team for the SAEINDIA AeroTHON – Rotorcraft Systems Challenge (UAS) – Design, Build and Fly Contest 2026. However, each team entry must present a unique and independently developed UAS design that is significantly different from the others.

If multiple teams from the same university or institution submit designs that are **not significantly different**, as determined by the **Organizing Committee**, the university/institution will be considered to have **only one valid entry**. In such cases, only **one team will be permitted to participate in the contest**.
For example, two designs having the **same configuration, motor layout, and overall dimensions** will not be considered significantly different.

Each team is expected to have a Faculty Advisor from the registered university or institution. Non-faculty members are not allowed to be advisors. The Faculty Advisor will be considered as the official university representative for that team by contest organisers.

Faculty Advisors may advise their teams on general engineering and engineering project management theory but should not be directly involved in the design of any part of the vehicle nor directly participate in the development of any documentation or presentation. They may review the design reports and provide suggestions and guide the team prior to the report submission and flying competition.

2.0. UAS DESIGN AND FLIGHT EQUIREMENTS

The objective of this year's contest is to design, build, and operate a multirotor Uncrewed Aircraft System (UAS) capable of successfully completing two mission scenarios that evaluate manual piloting skills, autonomous navigation, sensing capability, and payload delivery performance.

In **Mission 1 – Eyes in the Sky: Obstacle, Observation & Payload Delivery (Fully Manual Mission)**, teams must manually pilot their UAS through a defined obstacle course while maintaining stable flight and avoiding restricted flying zones. During the mission, the UAS must reach designated observation zones where specific observation tasks will be performed. Upon successful completion of the obstacle course and observation tasks, the UAS must deliver a payload accurately to a designated target area.

In **Mission 2 – SkyScan: Autonomous Rapid Delivery (Fully Autonomous Mission)**, the UAS must operate in a fully autonomous mode. The drone must autonomously take off, scan a QR code, navigate through a corridor, and locate the delivery zone. The system must identify the correct QR code associated with the delivery target and avoid restricted **red zones** within the mission area. Once the correct location is identified, the UAS must deliver the payload accurately by lowering it to the target point maintaining mission altitude. After completing the delivery, the UAS must autonomously return to the take-off point by navigating back through the corridor while avoiding obstacles, without any manual intervention.

Teams are required to design their UAS with the capability to **carry and deploy the payload securely, perform stable manual flight through obstacles, and execute autonomous navigation, sensing, QR code detection, and precision payload delivery** to successfully complete both missions.

2.1 DESIGN REQUIREMENTS

The design requirements of the UAS are listed in Table 2 and the payload dimensions for missions 1 and 2 are shown in Figure 1.

Table 1: UAS Design Requirements

Sl. No.	Parameter	Requirement/Limitation
1.	UAS Type	Multirotor
2.	UAS Category	Micro UAS (i.e., Take-off weight < 2kg)
3.	UAS MTOW	2 kg
4.	Payload Capacity	100 Grams
5.	Propulsion Type	Electric
6.	Communication System Range	At least 1 km

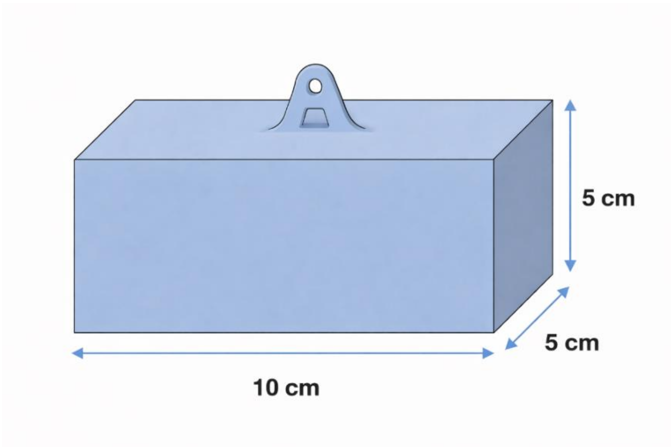

Figure 1: Payload Dimensions – Missions 1 & 2

Note: AeroTHON 2026 Track 1 contest is only for multirotor UASs. Fixed wings and VTOL Fixed wings are not allowed. Students are expected to bring innovation in the payload dropping method and mechanism to ensure a safe and precision delivery of payload to the target point. Provide design and analysis details of various systems and sub-systems, selection of Commercially Off the Shelf (COTS) items like batteries, motors etc. Students should consider safety of the platform and the environment in the design and highlight the risks and how they have been mitigated in the design.

3.0. PHASE 1: TECHNICAL DESIGN REPORT & PRESENTATION

In Phase 1, participating teams are required to submit a comprehensive Technical Design Report describing their proposed Unmanned Aerial System (UAS). The report must strictly adhere to the design requirements, mission objectives, and constraints specified in Section 2.0 of this rulebook. Teams should clearly demonstrate the technical feasibility, design methodology, innovation, and operational efficiency of their proposed system.

The design report should include detailed explanations of the **airframe configuration, propulsion system, avionics architecture, payload integration, control strategy, safety considerations, and mission execution approach**. Teams are also encouraged to justify their design decisions through **analysis, calculations, simulations, or experimental validation wherever applicable**.

In addition to the written submission, shortlisted teams will be required to **present their design to a panel of industry professionals and academic experts**. The presentation will provide teams with an opportunity to explain their design approach, highlight innovative aspects, and address questions from the evaluation panel.

The **Phase 1 evaluation** will consider the following aspects

- Technical depth and engineering rigor
- Innovation and originality of the design
- Compliance with the contest requirements and constraints
- Feasibility of mission execution
- Quality of analysis, documentation, and presentation

Only teams demonstrating a **strong technical foundation and a viable design** will be selected to advance to the **next phase of the competition**.

3.1 TECHNICAL DESIGN REPORT

The **Design Report** serves as the main tool for teams to communicate to the judges how their Uncrewed Aircraft System (UAS) was designed to best accomplish the intended mission. The report should outline the team's design decisions, demonstrating how the UAS is specifically suited to perform the required tasks. It should explain the team's thought processes and engineering philosophy that guided their choices, along with a detailed description of the methods, procedures, and calculations used to reach the final design solution.

Teams are required to submit the Design Report as per the timeline given in Section 1.6 and prepare a detailed presentation (Microsoft Power Point Format) and present it to the jury panel. The design report and presentation must have the following contents:

Technical Design Report Contents

The Technical Design Report (TDR) submitted in Phase 1 must clearly document the conceptual development, engineering analysis, and final design of the proposed UAS. The report shall include, but not be limited to, the following sections.

a) Conceptual Design

- **High-Level Physical Overview**
  Provide a detailed description of the primary physical elements of the UAS and their arrangement, illustrating the overall design approach.

b) Detailed Design

- **Preliminary Weight Estimation**
  Estimate the initial weight of the UAS, considering all components and their contributions.
- **Thrust Requirements Estimation**
  Calculate the thrust required for optimal UAS performance based on design specifications.
- **Propulsion System Selection**
  Select and justify the choice of propulsion system (e.g., motors, engines), ensuring compatibility with the overall design.

- **UAS Sizing**

  Provide detailed sizing parameters for key components, including:

  - Wheelbase
  - Rotor arm length
  - Hub size
  - Propeller clearance
  - Landing gear dimensions

- **UAS Performance Estimation**

  Estimate critical performance metrics, including:

  - Power requirements
  - Battery selection
  - Endurance estimation

- **Material Selection**

  Justify the choice of materials for the UAS structure, considering weight, strength, and durability.

- **Subsystem Selection**

  Identify and justify the selection of key subsystems, such as:

  - Communication system
  - Control & navigation systems
  - Avionics and sensors

- **Center of Gravity (C.G.) Estimation & Stability Analysis**

  Calculate the center of gravity and conduct a stability analysis to ensure proper flight dynamics.

- **Preliminary CAD Model**

  Teams shall submit preliminary 2D engineering drawings (front, top, and side views) along with a 3D CAD model of the proposed UAS configuration.

- **Computational Analysis**

  Conduct computational analysis to verify design performance and efficiency, including simulations.

- **Optimized Final Design**

  Present a summary of any design changes or optimizations made, with:
  - Final CAD model
  - Final 2D drafting
  - Updated C.G. analysis
- **Detailed Weight Breakdown & C.G. Analysis**

  Provide a detailed breakdown of the final UAS weight distribution and a thorough center of gravity analysis.
- **UAS Performance Recalculation**

  Recalculate performance metrics including:
  - Thrust-to-weight ratio
  - Power requirements for the mission
  - Endurance calculation

c) Final UAS Specifications and Bill of Materials (BoM)

- Submit a detailed specification of the final UAS design, along with an exhaustive Bill of Materials (BoM), listing all components used in the construction of the UAS.

d) Methodology for Autonomous Operations

Teams shall provide a detailed methodology for implementing the autonomous mission, addressing the following aspects:

- QR Code Detection and Identification - Describe the approach and algorithms used for detecting and decoding QR codes during the mission. Teams should explain how the system identifies the relevant QR code information required for navigation and delivery.
- Autonomous Navigation Through the Corridor - Explain the strategy used for autonomous flight through the designated corridor, including navigation control, obstacle detection, and path planning to safely traverse the corridor without manual intervention.

- Delivery Zone Identification - Describe the method used to locate and identify the correct delivery zone based on the QR code information obtained during the mission.
- Restricted Zone Avoidance - Explain the techniques used to detect and avoid restricted (red) zones during flight while navigating toward the delivery location.
- Payload Delivery Methodology - Describe the payload delivery mechanism and control logic used to lower and release the payload accurately at the designated target location.
- Autonomous Return and Landing Strategy - Explain the strategy for the autonomous return flight to the take-off location, including safe navigation back through the corridor while avoiding obstacles.

Summary of Innovations

- Provide a concise summary of the innovative aspects of the proposed UAS design. Teams should highlight any **novel features, advanced technologies, unique design approaches, or system integrations** that distinguish their design from conventional solutions and enhance the overall mission capability.
- If applicable, teams should also explain how these innovations **improve performance, reliability, efficiency, or mission effectiveness**.

Refer to Sections 5.0, 6.0, and 7.0 for detailed information on the evaluation criteria, Technical Design Report submission requirements, and presentation guidelines.

3.2 DESIGN PRESENTATION

In addition to the Technical Design Report, teams are required to present their proposed UAS design to a panel of jury members comprising industry experts, subject matter specialists, and academic professionals.

The presentation should clearly communicate the **key elements of the team's design process**, demonstrating how the proposed UAS meets the **mission objectives and competition requirements**. Teams are expected to explain their **design methodology, engineering analyses, subsystem selections, and the rationale behind major design decisions**.

The presentation should be structured in a manner that effectively engages an audience with diverse technical backgrounds, ensuring that the content is **technically sound while remaining clear and comprehensible**. The use of **visual aids such as slides, diagrams, CAD models, simulations, or system architecture diagrams** is strongly encouraged to support the explanation of the design and technical concepts.

Teams must also be prepared to **respond to questions from the jury panel and justify their design decisions**, demonstrating a thorough understanding of the engineering principles and trade-offs involved in their UAS design.

4.0. PHASE 2: FLYING COMPETITION

Phase 2 of the SAEINDIA AeroTHON – Rotorcraft Systems Challenge (UAS) – Design, Build and Fly Contest 2026 will consist of two stages:

1. Technical Inspection
2. Flying Competition

4.1 TECHNICAL INSPECTION

All Uncrewed Aircraft Systems (UAS) will undergo a **Technical Inspection** conducted by designated UAS technical inspectors before being permitted to participate in any flight demonstrations. The inspection will assess compliance with both technical and safety standards in accordance with general industry safety guidelines. The decisions of the UAS inspector(s) are final.

The **Technical and Safety Inspection** will verify the following:

1. **Compliance with UAS Design Requirements:**
   - Ensuring adherence to the specified UAS design criteria as outlined in Phase 1
2. **Overall Safety and Airworthiness:**
   - Evaluation of the UAS's structural integrity, safety features, and flight readiness.

All UASs must successfully pass the **Technical and Safety Inspection** to be eligible for the flying competition. Teams are strongly encouraged to perform a self-inspection prior to arrival at the contest to ensure compliance.

During the **Technical Inspection**, the following checks will be conducted:

- **UAS Dimensions:** Verification that the UAS conforms to the 2D drawings submitted during Phase 1.
- **Component Verification:** Ensuring the use of components as specified in Phase 1, including:
  - Propulsion system: Motor, Electronic Speed Controller (ESC), Propeller, and Power System (Battery)
  - Control & Communication System: Flight Controller, Radio Transmitter, and Receiver

- **Take-off Weight:** Confirmation that the UAS weight matches the value reported in the design report.
- **Structural Integrity:** Examination of the UAS's structure, including:
  - Proper securing of components, appropriate wiring (no hanging wires, use of proper wire gauge and connectors)
  - Secure fasteners (locknuts or thread locker used, no loose or shaking components)
  - Propeller and payload attachment integrity
- **Other Checks:**
  - Control system responsiveness (motor RPM) to radio controller inputs
  - Motor/propeller rotation direction
  - Radio range check
  - Motor arming and disarming check
  - FPV video transmission check
  - Any deviations from the above checks must be addressed through the Change Request Process as outlined in APPENDIX B

A detailed technical inspection will be conducted prior to the first flight, while subsequent flights will be subject to a visual inspection, provided the UAS remains incident-free. If the UAS is damaged during a mission or trial, a detailed inspection will be performed again. It is recommended that teams bring strategically selected spare parts and components in the event of any unforeseen incidents during transit or the competition.

4.1.1 UAS Conformance to 2D Drawing

During Technical Inspection, the UAS will be inspected and measured for conformance to the 2D drawing presented in the Design Report.

a) At a minimum, UAS arm length, landing gear height and UAS height dimensions will be measured and compared to the 2D drawing.
b) All teams must have a hard copy of their design report with them during technical inspection.
c) UAS actual empty CG will be compared to the empty CG presented in the design report's 2D drawing.

4.1.2 Deviations from 2D Drawing

Any deviation in construction of the UAS from the submitted 2D drawing since submission of the Design Report must be reported in writing.

- Each design change must be documented separately using the Modification Change Request (CR).
- Only one design change may be submitted by CR form.
- Jury will assess penalty points for design changes.

4.1.3 Inspection of Spare UAS Components

- All spare UAS components (structural parts, motors, propeller, batteries etc.) must be presented for inspection at the same time as the UAS inspection.

4.1.4 Inspection Requirements throughout the Contest

- All UAS must meet all Technical and Safety Inspection requirements throughout the contest.
- Any official may request that an UAS be re-inspected if a general or safety requirement problem is seen on an UAS at any time during the event.
- This includes any unintended errors or omissions made by officials during inspection.

4.1.5 Technical and Safety Inspection Penalties

- Points are allotted for the Technical and Safety Inspection.
- Teams may only lose points because of errors and problems encountered during the inspection process. Any penalties assessed during Technical Inspection will be applied to the overall contest score.

4.2 FLYING COMPETITION

Participants shall refer to the SAEINDIA AeroTHON 2026 website https://saeindia.org/aerothon2026/ for the Phase 2 flying competition dates and schedule.

4.2.1 General Mission Requirements

The objective of **AeroTHON 2026** is to design, build, and operate a multirotor **Uncrewed Aircraft System (UAS)** capable of completing manual obstacle navigation and observation tasks, as well as performing autonomous QR-based navigation and precise payload delivery

Aircraft Consistency:

- The aircraft must remain substantially identical to the design documented in the submitted report.

Timely start:

Teams must report to the assigned flight field strictly as per the scheduled time slot. If a team is not present at the field or is unable to initiate take-off within 5 minutes of being called to the podium, a penalty will be imposed for the delay. The nature of the penalty will be communicated by the organizers and may range from negative scoring to forfeiture of the mission. **Failure of a team to report during their allocated time slot will be treated as a forfeiture of that mission attempt**, and the team will not be permitted to perform the mission later unless otherwise decided by the organizing committee.

Flight Missions:

The competition will consist of two distinct flight missions:

- **Mission 1:** Eyes in the Sky – Obstacle Navigation, Observation & Precision Payload Delivery – Manual Operation
- **Mission 2**: SkyScan – Autonomous QR Detection, Corridor Navigation & Rapid Payload Delivery – Autonomous Operation

Scoring:

- Scoring will be awarded individually for each mission: one for manual operation and one for autonomous operation.

Damage and Repairs:

- If the UAS is damaged after an unsuccessful flight mission, teams may carry out necessary repairs, provided that no modifications are made that deviate from the design submitted in Phase 1. However, the UAS must undergo a re-inspection and be cleared as airworthy before the next mission attempt. Failure to pass the inspection will result in disqualification from the next mission attempt.

In the Event of Irreparable Damage:

- If the UAS is damaged beyond repair or deemed unfit for flight by the UAS inspectors, the team will forfeit their opportunity to attempt the next mission.

Design Deviations:

- Any deviation from the original design submitted in Phase 1 during repairs or overhauling will be grounds for disqualification, subject to the decision of the Jury Panel.

4.2.2 Flying Competition Chronology

- The flying schedule will be sent via email to all teams at least two days ahead of the demonstration event, also a copy of schedule will be shared with the Teams during Phase 2 event registration. Teams are expected to follow this schedule during their flight presentations.

  The schedule must be adhered to without exception. Teams that do not attempt the flying event at their designated time will get penalty and opportunity would be given upon Technical Committee advise.

- Each team will have a 15-minute window to complete their mission. If a team encounters issues during their scheduled attempt due to the following reasons, they can overhaul the UAS and try again, if their initial flight time is under 2 minutes:
  - UAS failed to take off
  - UAS took off but landed immediately
  - UAS took off and crashed

Note: Jury members hold the authority to make final decisions in these cases

Inability to Engage in Phase 2 Scenarios

- If any team is unable to attend the flying competition due to conflicts with examination schedules, medical emergencies, or similar reasons, a delegate or substitute may be permitted to fly the UAS at the venue, provided a letter is issued by the Institute's HoD, Principal, or Faculty Advisor.
- The delegate or substitute must be from the same university but does not have to be from the same team.
- Teams must notify SAEINDIA in writing, along with the letter, at least 10 days in advance.

- The SAEINDIA AeroTHON 2026 Organizing Committee reserves the right to approve requests for delegates or substitutes. The committee's decision will be final and binding for all teams.

4.2.3 Flight Mission 1 – Eyes in the Sky: Obstacle, Observation & Payload Delivery – Manual Operation

Mission Objective

- Navigation through obstacles: transport a payload through a challenging course filled with static obstacles such as hurdles, tunnels, hoops, and slalom sections while avoiding restricted or no-Fly zones. The primary objective is to navigate these obstacles with high precision while ensuring the payload remains undamaged.
- Arial Survey: Continue the course passing through an observation zone, identify and report identified visual observations or objects. This phase of the mission is typically to demonstrate applications of drones in various segments e.g. Agri, field survey, remote area surveillance and monitoring etc.
- Payload Delivery: Deliver payloads in identified zone post in-flight navigations through obstacles and avoiding no-fly zones. This phase is to replicate typical urban delivery applications

Mission Details:

• In this mission, the UAS must be **manually piloted through a structured obstacle course** consisting of various challenges such as hurdles, tunnels, hoops, and slalom sections. The objective is to demonstrate **precise piloting skills, situational awareness, and stable flight control** while carrying the payload.

• The mission course may also include **restricted or no-fly zones** marked with red banners or boundary markings. Pilots must carefully navigate the drone to avoid entering these restricted areas while continuing the mission path.

• During the mission, the drone must also perform **observation tasks at designated observation zones**. These tasks may include identifying visual elements such as **aerial views of structures or infrastructure, or 2D/3D shapes** placed within the course. Teams must **hover at the observation zone and verbally report the observation to the jury** before continuing the mission.

• After completing the obstacle navigation and observation tasks, the drone must proceed to the **payload delivery zone** and accurately place the payload within the designated target area.

• The mission is considered complete when the drone successfully delivers the payload and returns home.

Mission Tasks:

- **Manual Obstacle Navigation:** Fly the drone through various obstacles such as hurdles, hoops, tunnels, and slalom gates while maintaining stable control.
- **Restricted Zone Avoidance:** Detect and avoid marked red/restricted zones without entering the restricted areas.
- **Observation and Reporting:** Hover at designated observation zones and correctly identify and verbally report the observed objects or features to the jury.
- **Payload Transport:** Safely carry the payload throughout the mission without dropping or destabilizing it during flight.
- **Precision Payload Delivery:** Place the payload accurately within the designated delivery zone.
- **Return to Home:** After the payload is placed, navigate the course back to the home base or takeoff point, avoiding obstacles and ensuring a safe return.
- **Time Constraints:** Complete the mission within a 15-minute time slot.

Operation:

- **Takeoff and Start:** The UAS must take off from the designated **start zone**. Pilots must remain within the **take-off control zone for the entire mission**.
- **Obstacle Course Navigation:** The pilot must manually maneuver the UAS through the obstacle course while maintaining stable flight and avoiding collisions.
- **Restricted Zone Avoidance:** Restricted areas marked by **red banners or ground markings** must not be entered. Pilots must adjust their flight path in real time to avoid these zones.
- **Observation Zones:** At designated observation points, the pilot must stabilize the UAS, visually identify the observation target, and **verbally report the observation to the jury** before continuing.

- **Payload Delivery:** After completing the required navigation and observation tasks, the pilot must proceed to the delivery zone and **place the payload inside the designated target area**.
- **Finish Zone:** The mission concludes once the drone **successfully delivers the payload** and UAS must return to the takeoff point and land safely.
- **Manual Control Requirement:** This mission must be performed **entirely under manual control**, and no autonomous flight assistance is permitted.
- **Safety Compliance:** Safety remains the highest priority. **Unsafe flying, repeated collisions, or violation of safety rules may lead to penalties or disqualification**.
- **Scoring:** A successful flight requires completing the obstacle course, placing the payload, returning to the home base, and landing safely. See Section 5.5 for scoring details.
- **Flight Time:** Teams have 15 minutes to complete the mission, starting when throttle input for takeoff is increased.
- **FPV Option:** Teams may use a First-Person View (FPV) camera to aid navigation, but it’s not mandatory.
- **Field Conditions:** Actual conditions may vary slightly from those depicted in Figure 2.

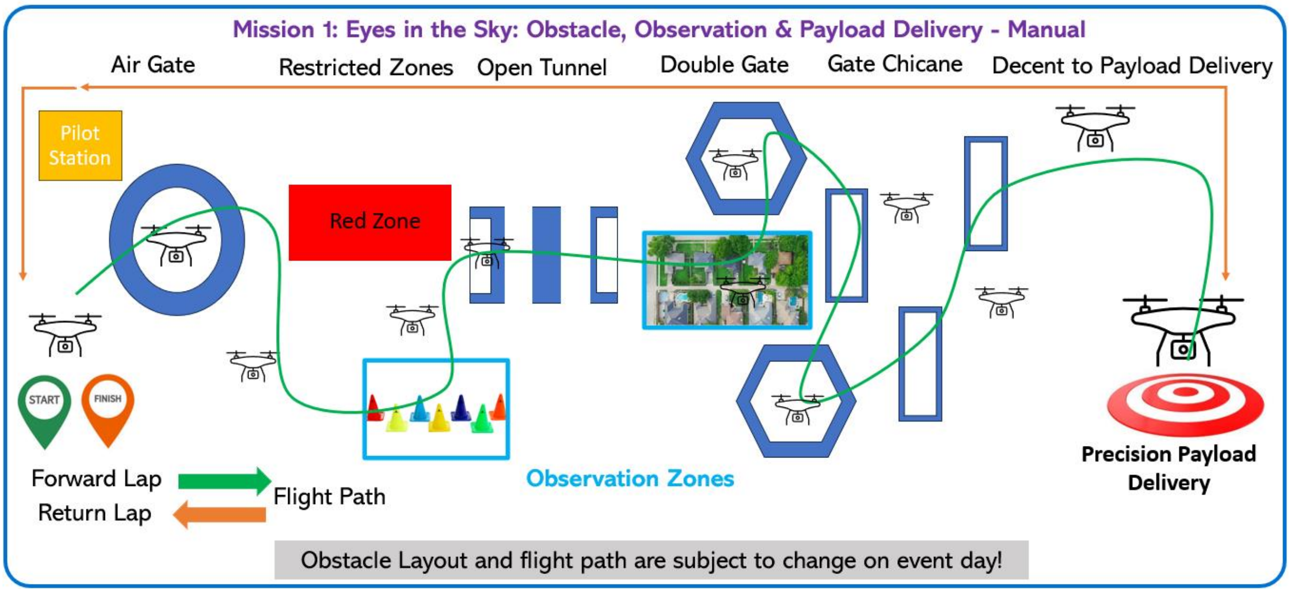

Figure 2 - Flight Mission 1 – Manual Mission Profile

The flight path will be clearly marked on the ground to assist the pilot in navigating the obstacle course and accurately delivering the payload. Additionally, a debrief will be provided before the mission commences.

4.2.4 Flight Mission 2 – SkyScan: Autonomous Rapid Delivery – Autonomous Operation

Mission Details:

• In this mission, the UAS must operate in **fully autonomous mode** to perform a rapid delivery task based on information obtained from a **QR code**. The drone must first scan the QR code to determine the required delivery location and then navigate through a designated corridor while autonomously avoiding obstacles.

• After scanning the QR code, the drone must descend and travel through a **3.5-meter-wide navigation corridor**, demonstrating its ability to perform **precise autonomous navigation avoiding the walls.**

• Upon reaching the delivery zone, the drone must ascend to the required altitude and identify the **correct QR code corresponding to the delivery location**. Multiple QR codes will be placed within the delivery zone, and the UAS must correctly identify the matching code while avoiding **designated restricted (red) zones**.

• Once the correct target location is identified, the drone must **descend to the specified altitude and lower the payload accurately** at the designated QR code location.

• After successfully delivering the payload, the drone must **ascend and autonomously return to the starting point**, navigating back through the corridor while avoiding obstacles, and land safely at the take-off location.

Mission Tasks:

- **QR Code Detection:** Take off and scan the start QR code from a 5-meter altitude and decode the delivery location information.
- **Corridor Entry Detection Forward Lap:** Detect the **AeroTHON 2026 Banner with Green Background** placed at the corridor entrance and autonomously align with the corridor before beginning corridor navigation.
- **Autonomous Corridor Navigation:** Navigate through a **3.5-meter-wide corridor** while autonomously avoiding obstacles.
- **Target Identification:** Ascend to 10-meter altitude and identify the correct QR code within the delivery zone that corresponds to the scanned delivery information.
- **Restricted Zone Avoidance:** Detect and avoid designated **red or restricted zones** within the delivery area.
- **Payload Delivery:** Descend to 5-meter altitude and accurately **lower and release the payload** at the correct QR code location.

- **Corridor Entry Detection Return Lap:** Detect the **AeroTHON 2026 Banner with Green Background** placed at the corridor entrance and autonomously align with the corridor before beginning corridor navigation.
- **Autonomous Return:** After delivery, return through the corridor while autonomously avoiding obstacles and land safely at the take-off point.

Operation:

- **Takeoff:** The UAS must take off autonomously from the designated ground start point.
- **QR Code Scan:** The drone must ascend to **5-meter altitude**, move forward approximately **1-meter**, and scan the QR code containing the delivery information.
- **Corridor Navigation:** After scanning the QR code, the UAS must identify the **AeroTHON 2026 Green Banner** marking the entrance of the navigation corridor and align itself before descending to the corridor navigation altitude, the drone must **descend to 10 feet (approximately 3-meter) altitude** and autonomously navigate through the corridor while avoiding obstacles.
- **Delivery Zone Navigation:** Upon reaching the delivery zone, the drone must **ascend to approximately 10 meters altitude** to identify the correct QR code. Delivery zone **geo-fence coordinates will be provided to teams during Phase 2**.
- **Payload Delivery:** Once the correct QR code location is identified, the drone must **descend to 5-meter altitude**, lower the payload, and release it accurately at the designated location.
- **Return Flight:** After payload delivery, the UAS must **ascend to 10-meter altitude**, navigate back through the corridor while avoiding obstacles, and return to the take-off location.
- **Landing:** The mission concludes once the drone lands safely at the designated take-off point.
- **Geo-fencing:** Coordinates for the geo-fence boundary will be provided. Teams must program these into the ground station software to ensure the UAS stays within the designated area.
- **Flight Time:** Each team will have 15 minutes to complete the mission. Scoring will be based on mission success.

Criteria for a Successful Flight:

A flight will be deemed successful only if the UAS:

- A flight will be considered successful only if the UAS:
- Takes off autonomously from the start point.
- Scans and correctly interprets the QR code.
- Successfully navigates the corridor while avoiding obstacles.
- Identifies the correct QR code within the delivery zone.
- Delivers the payload accurately at the designated location.
- Returns through the corridor and land safely at the take-off point. Please refer to Section 5.6 for detailed scoring criteria. The course for autonomous operations is shown in Figure 3; however, field conditions may vary.
- The UAS should be equipped to record flight data and share it with the jury panel once the mission is completed.

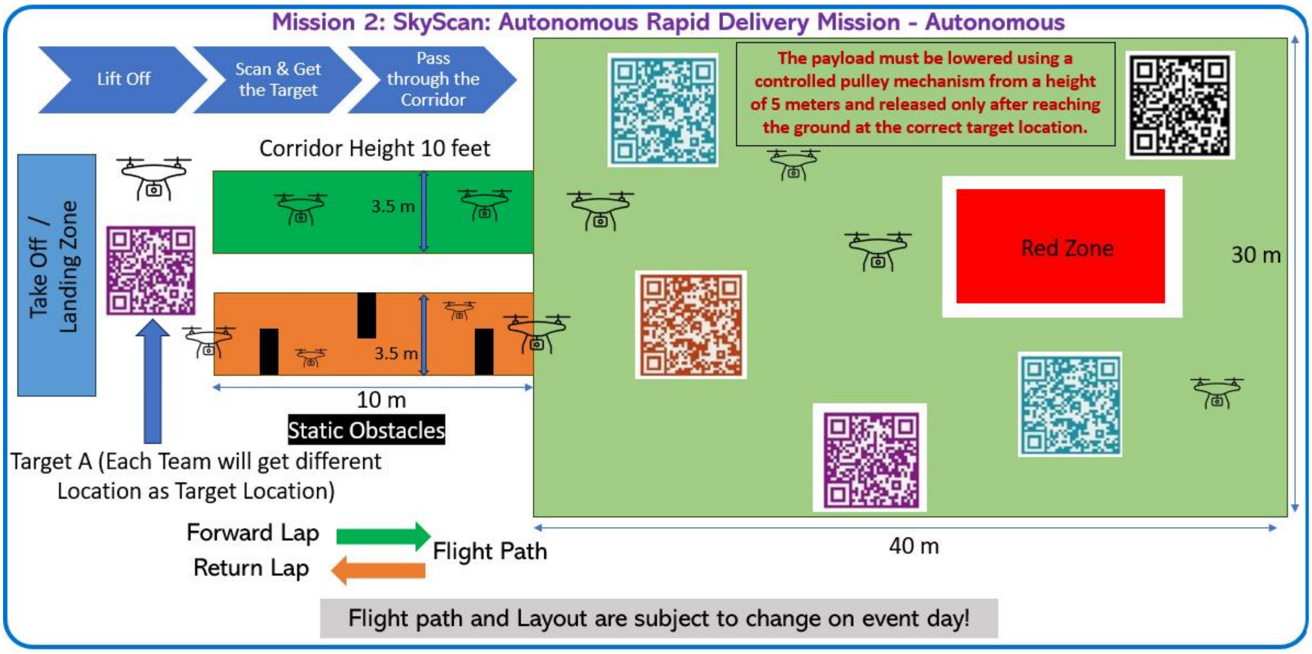

Figure 3 - Flight Mission 2 – Autonomous Mission Profile

5.0. EVALUATION CRITERIA

5.1 PHASE 1 - DESIGN REPORT SUBMISSION

PHASE 1 - DESIGN REPORT EVALUATION – 50 Marks		
Parameter		Score
1	Technical Content	20
1.1	Conceptual Design Process	2
1.2	Product Benchmark	2
1.3	Preliminary Weight Estimation	2
1.4	Thrust Required Estimation & Propulsion System Selection	2
1.5	Aircraft Sizing (Rotor Arm, Hub, Wheelbase, Propeller Clearance, Landing Gear)	2
1.6	Aircraft Performance (Power Estimation, Battery, Endurance)	2
1.7	Material Selection	2
1.8	Subsystem Selection (Communication, Control, Navigation)	2
1.9	Autonomous Navigation System Hardware & Software	2
1.10	C.G. Calculation & Stability Analysis	2
2	Computational Analysis	6
Computational Analysis (CFD, FEM, MATLAB, etc.)		6
3	Methodology for Autonomous Operation	12
3.1	Autonomous Flight Algorithm.	4
3.2	Autonomous Object Detection and Counting Algorithm	4
3.3	Autonomous Payload Drop Mechanism including algorithm, hardware, and system selection	4
4	Flight Safety	2
4.1	Safety Features & SORA Assessment	2
Assesses the implementation of flight safety features, such as Return-to-Home, battery fail-safe, telemetry fail-safe, and radio fail-safe		
5	Innovation	10
Participants must bring innovative solutions to address challenges in the drone industry, such as battery life, safety, and regulations. Innovation should span the UAS life cycle—design, manufacturing, testing, operations, and maintenance—focusing on efficiency, autonomy, and sustainability. Solutions should leverage advanced materials, AI, and smart systems to enhance performance and overcome current limitations.		

5.2 PHASE 1 - TECHNICAL PRESENTATION

PHASE 1 - TECHNICAL PRESENTATION – 50 Marks		
Parameter		Score
1	Technical Design & System Architecture	12
Overall UAS design, structural configuration, propulsion system selection, payload integration, stability considerations, and alignment with mission requirements		
2	Mission Strategy & System Integration	10
Clear explanation of how the UAS design supports both missions' including obstacle navigation, observation tasks, QR scanning, corridor navigation, payload lowering mechanism, and return strategy.		
3	Autonomous Operation Methodology	10
Algorithms and approach for QR code detection, corridor navigation, obstacle avoidance, red-zone avoidance, delivery zone identification, and autonomous payload deployment.		
4	Engineering Analysis & Design Validation	8
Weight estimation, thrust-to-weight calculations, endurance estimation, CG analysis, safety margins, and simulation or computational validation of the design.		
5	Structure & Clarity of Presentation	5
Logical flow of ideas, clear introduction and conclusion, effective use of diagrams, CAD visuals, and ability to explain complex ideas clearly within the time limit.		
6	Presentation Skills & Technical Communication	5
Clear, confident communication, teamwork, ability to answer technical questions, interaction with judges, and professionalism during the presentation.		

5.3 PHASE 1 - TOTAL SCORE

PHASE 1 - TOTAL SCORE – 100 Marks		
EVENTs		Score
1	**TECHNICAL DESIGN REPORT**	50
2	**PRESENTATION**	50
	TOTAL SCORE	100

5.4 PHASE 2 - TECHNICAL INSPECTION – 50 Marks

PHASE 2, TECHNICAL INSPECTION – 50 Marks		
Parameter		Score
1	Aircraft Dimensions Conformance to 2D Drawings (10)	
1.1	Verification of Dimensions against Phase 1 design submitted values	10
The design of the frames and their dimensions should strictly adhere to the Phase 1 design specifications. There will be significant penalty points if the UAS does not conform to the Phase 1 design, potentially leading to disqualification.		
2	Use of Same Components as Selected in Phase 1 with identical technical specifications (6)	
2.1	Propulsion System: Motor (KV Rating), ESC (Amps), & Propeller (Size)	2
2.2	Power System: Battery [Voltage (V), Capacity (mAh), & C-Ratings]	2
2.3	Control & Communication System: Flight Controller, Radio Transmitter & Receiver	2
3	Take-off Weight Consistency with Design Report (10)	
3.1	Weight Difference < 50g	10
3.2	Weight Difference > 50g & < 100g	8
3.3	Weight Difference > 100g & < 200g	5
3.4	Weight Difference > 200g	0
4	Structural Integrity (10)	
4.1	Components Secured Properly	2
4.2	Proper Wiring (No wires hanging, use of appropriate gauge wires and connectors)	2
4.3	Secure Fasteners (use of locknuts, thread locker)	2
4.4	Proper Payload Attachment	2
4.5	No Loose or Shaking Structural Components	2
5	Static System Checks (8)	
5.1	**Motor & Propeller Check**: Inspect mounting and ensure no visible damage.	2
5.2	Battery Inspection (secure attachment, no visible damage, proper wiring)	2
5.3	**Controller System Check**: Ensure the flight controller is secure and wiring is intact.	2
5.4	**Radio Range Check**: Move the transmitter to a set distance to test UAS communication.	2
6	Fail-Safe Systems Check (4)	
6.1	**RTL on Low Battery**: Ensure low battery return-to-launch functionality is configured.	1
6.2	**RTL on Datalink Loss**: Ensure automatic return is configured in case of data loss.	1
6.3	**Geo-Fence Check: Verify geo-fence limits and configurations.**	2
7	Aesthetics (2)	
7.1	**Inspector Marks**: Based on build quality, attention to detail, and system integrity.	2

5.5 PHASE 2 - FLIGHT MISSION 1- Manual Operation – 100 Marks

PHASE 2 - FLIGHT MISSION 1 – 100 Marks		
	Parameter	**Score**
1	Obstacle Navigation	40
UAS successfully takes off and maintains the required mission altitude throughout the obstacle navigation course. UAS skillfully navigates through the obstacles on the forward lap without causing damage to the UAS or obstacles. Points are awarded based on precision and smoothness. Penalties are applied for collisions or crashes, with up to Negative 5 points subtracted for each incident.		
2	Red /Restricted zone avoidance	15
UAS shall avoid identified no-fly zones during navigation. Points are awarded for successful avoidance of identified red zones. Each violation to avoid the Red /Restricted zones could lead to **negative 10 points** subtraction from the scores earned		
3	Arial Survey: In – flight Observation	15
UAS to have required systems and navigate observation zones to perform in flight arial survey of identified Observation Zone(s). Points are awarded for successfully identifying and manually reporting to the jury about the objects identified. A **negative 5 points** shall be avoided for each incorrect object identification.		
4	Payload Delivery	10
Points are awarded for identifying and successfully placing the payload at Ground Zero (target point). The more accurate the placement, the higher the score.		
6	Take-off, Return to Take-off Point & Land Safely without functional damage	5
The UAS returns to the takeoff point and lands safely and no functional damage to UAS		
7	Within Time Completion & Start	15
The UAS completes the mission within the **allotted 15-minutes** flight time.		

Note:

The above scoring criteria are preliminary and may be updated for the Phase 2 Flying Competition. Final evaluation criteria will be shared after Phase 1 to ensure fair and transparent assessment.

5.6 PHASE 2 - FLIGHT MISSION 2 - Autonomous Operation – 100 Marks

PHASE 2 - FLIGHT MISSION 2 – 100 Marks		
	Parameter	**Score**
1	Autonomous Takeoff and Initial QR Code Detection	10
The UAS autonomously takes off, ascends to approximately 5 m, moves forward and successfully scans and decodes the QR code containing the delivery location information. Marks proportional for partial success.		
2	Autonomous Corridor Navigation	15
The UAS descends to approximately 3 m (10 feet) altitude and successfully navigates through the 3.5-meter-wide corridor while maintaining stable autonomous flight. Marks proportional for partial completion or minor deviations		
3	Delivery Zone QR Code Identification	10
The UAS ascends to approximately **10 m altitude** and correctly identifies the **target QR code** corresponding to the scanned delivery information among multiple QR codes placed in the delivery zone. Partial marks for incorrect or delayed identification.		
4	Restricted Zone / Red Zone Avoidance	10
The UAS successfully detects and avoids designated red zones while navigating within the delivery area. Each violation to avoid the Red /Restricted zones could lead to **negative 5 points** subtraction from the scores earned		
5	Payload Delivery Accuracy	15
The UAS descends to approximately **5 m altitude**, lowers and releases the payload accurately at the **designated QR code location**. Marks will be awarded based on drop accuracy, altitude compliance, and stability during delivery.		
6	Autonomous Return Navigation	20
After payload delivery, the UAS ascends and autonomously navigates **back through the corridor**, avoiding obstacles while returning to the start point.		
7	Safe Autonomous Landing	5
The UAS lands safely at the designated take-off location **without manual intervention.**		
8	Mission Completion Within Time	15
The UAS completes the mission within the allotted 15-minutes flight time. The marks will be proportional to how quickly the mission is completed.		

Note:

The above scoring criteria are preliminary and may be updated for the Phase 2 Flying Competition. Final evaluation criteria will be shared after Phase 1 to ensure fair and transparent assessment.

5.7 PHASE 2 - TOTAL SCORE

PHASE 2 - TOTAL SCORE – 100 Marks		
EVENT / MISSIONs		Score
1	TECHNICAL INSPECTION	50
2	FLIGHT MISSION - 1	100
3	FLIGHT MISSION - 2	100
	TOTAL SCORE	250

5.8 TIGER'S CAVE- BUSINESS PLAN PROPOSAL

This year AEROTHON organising committee has included a new stage specially for developing entrepreneur mindset. Through this stage, we are looking for teams that have an appetite to not just solve challenges but also set new trends of growth and success in Drone industry.

During this event, teams would pitch their product in front of notable jury from successful drone start-ups and related industries and gain insights from them. This opportunity to pitch your product in front of industry trailblazers is a win, the best team will be awarded a special category prize.

The team will be presenting to a panel of 3-4 tigers with a time limit of 15 minutes (10 mins presentation & 5 mins Q&A). The team is free to decide on the format of their presentation, props, display parts to make an impact.

The evaluation criteria are not defined as the winning team decided by the tigers, but the recommended topics to cover could be:

- Business plan proposal
- Capability of your UAS- key points
- Key Technical aspects and differentiators
- Practical market/ use case to which your system could be implemented.
- Market survey on the scope and growth of your use case
- Unique Selling Point (USP) of your design.
- Innovations used in your product which sets it apart from other teams.
- Financial, fund raising, sponsorships so far, plan to scale up.
- Team's structure for future success

These are just helpful points to start with, but the teams are free to decide what points to cover and present them. Think of it as a shark tank experience.

This is an imagination exercise of what your product could be!!

<u>Note:</u> Scores of this stage will not be included in overall contest score.

6.0. DESIGN REPORT GUIDELINES FOR PHASE 1

6.1 INTRODUCTION

Technical report writing is a skill that is different from informal writing – letters, notes, email – and, like all skills, needs the practice to master them. The SAEINDIA AEROTHON – ROTORCRAFT SYSTEMS CHALLENGE (UAS) – DESIGN, BUILD AND FLY CONTEST 2026 provides an excellent opportunity for students to exercise this skill. This document provides guidelines to help design teams write clear, concise, and data-rich reports.

6.2 ORIGINAL WORK

The Technical Design Report shall be the team's original work for the current contest year. Resubmissions of previous and current year's design reports will not be accepted. Recitation of previous year's work is acceptable if and only if appropriately cited and credited to the original author(s). Plagiarism is forbidden in industry and academic practice. All references, quoted text and reused images from any source shall have an appropriate citation within the text and within the Technical Design Report's table of references, providing credit to the original author and editor.

Reports may be checked against previous and current years' submissions to determine if re-use, copying, or other elements of plagiarism are indicated.

For the SAEINDIA AEROTHON – ROTORCRAFT SYSTEMS CHALLENGE (UAS) – DESIGN, BUILD AND FLY CONTEST 2026, plagiarism is defined as any of the following:

a) Use of information from textbooks, reports, or other published material without proper citation

b) Use of sections or work from previous SAE Aero Design contests without proper citation

If plagiarism is detected in the design report, the team will be disqualified, or points will be deducted as deemed by the rules committee/ jury depending on the amount of plagiarized content present in the design report.

The SAEINDIA AEROTHON – ROTORCRAFT SYSTEMS CHALLENGE (UAS) – DESIGN, BUILD AND FLY CONTEST 2026 Rules Committee & SAEINDIA has the sole discretion to determine whether plagiarism is indicated, and the above rules are enacted. The above rules may be implemented at any time before, during, or for up to six (6) months after the contest.

6.3 ORGANIZATION OF CONTENTS

Reports are written for a person or group to read, and these readers have a purpose for reading the report. In the SAE contest, the readers are the jury, and their purpose in reading is to grade the paper. Therefore, the design team authors should write the design report using techniques that make it easy for the jury to grade them. Organizing the report for the reader's purpose is the first technique for effective technical writing.

Outline - The judge's grading criteria predominantly depend on the technical aspects. So, the teams are expected to,

- Explain the team's thought processes and engineering philosophy that drove them to their conclusions.
- Detail the methods, procedures, and where applicable, the calculations used to arrive at the presented solution.
- Covering these topics
    - ❖ UAS configuration selection
    - ❖ UAS design includes rotor arm, hub, landing gear, etc.
    - ❖ Subsystem Selection
    - ❖ UAS Performance
    - ❖ UAS C.G., stability, and control
    - ❖ Computational Analysis
    - ❖ Other as appropriate

It also covers the administrative aspects of the report – page limits, formats, and specific graphs and drawings. Although it may be harder to write the report to this outline, it will be easier for the jury to grade it. This outline also forces the team to address topics the jury **must** grade and develop necessary data.

6.4 WRITING PROCESS

Writing a multi-page design report can be made less daunting by using a multi-step process. The first step is described above, generating an outline that addresses the reader's purpose. The next steps, described below, help in generating a data-rich, well-edited design report.

Allocate Pages – Allocate 40 pages to the sections of the outline. The allocations should reflect the emphasis on areas of the team's design. Do this before writing begins and adjust after reviewing the first draft. For each page of the design report, define the topic to be discussed and the message to be delivered. Make writing assignments for each page. Giving authors page-by-page assignments makes it easier to attack the writing – they are writing only one page at a time.

Create the Figures – Most juries will be engineers, and engineers are graphically inclined - they can understand a concept more easily when looking at a picture. Therefore, build each page around at least one figure. Create the figures first and review them before starting to write. Each figure needs a message which should be summarized in the figure title. Make the figures data-rich, but legible (9-point font is a minimum size - another advantage of using figures is that the rules do not constrain type font or spacing on figures). Equations can be incorporated in figures to save space.

Draft the Text - Use text to highlight, explain, or further develop the major points of the figure. Writing guidelines for clarity and succinctness are presented in a subsequent section.

Edit the Text and Figures – Take the time to edit the document at least twice. A good approach is to perform one edit cycle based on a group review of the draft document (called a Red Team). Have the Red Team members read the document as juries, supplying them with a scoring sheet and a copy of the rules.

Create the Final Document – Although several persons may contribute to the writing process, one team member should make the final version. This person works to achieve a consistent style to the text and to make the messages consistent.

Schedule the Effort – Although this is the first step, it is described last so that the reader can see what the team needs to schedule! A good report takes more than a week to create. One month is a guideline for the duration of the writing effort. Create a schedule of the above tasks and status it regularly. An efficient method is to establish the outline, page allocations, and figures early in the project, so the team can generate the necessary data as the design progresses. This reduces both the last-minute cram and the amount of unused documentation.

6.5 DESIGN REPORT SPECIFICATIONS

6.5.1 Page Limit

The design report must not exceed forty (40) single-spaced, typewritten pages, cover page, table of contents and appendix. The maximum limit of the document is given below:

Document	Max. Number of Pages
Main content	30
Appendix- additional supporting material	10

Note: Statement of Compliance will not be counted toward the 40-page limit.

6.5.2 Electronic Report Format

All reports must be submitted in (.PDF) format only. The document should be submitted electronically, and no handwritten documents will be accepted.

6.5.3 Font

The minimum font size and type is Arial 12 point proportional.

6.5.4 Margin

The report margins shall be: 1” Left, 0.5” right, 0.5” top, and 0.5” bottom. Each page, except the cover page, Certificate of Compliance, 2D Drawing and technical data sheet shall include a page number.

6.5.5 Page size

All report pages shall be A4 portrait format.

6.5.6 Cover page

All Design Reports must feature a cover page that states the team's name, college or university, and team number. The cover page will count against the 30-page limit.

6.5.7 Submission of Reports

Teams are required to upload technical reports to PDF file by the deadline date at the web link.

6.6 ELECTRONIC DOCUMENT SPECIFICATIONS

6.6.1 Format Size

Plan sheet must be in A3 page (PDF) format (11 x 17 inches). Plans must only consist of one (1) page and must have the US-standard third-order projection.

6.6.2 Required Views

The plans shall consist of a standard aeronautical three-view, using a US-standard third order projection, i.e., right side view in the lower left with the nose pointing right, top view above the right-side view also with the nose pointing right, and front view in the lower right.

6.6.3 Dimensions

At a minimum, the UAS must have length, width, height, and CG location marked clearly and dimensioned in the submitted engineering drawings. All dimensions must be in Metric units to an appropriate level of precision. (Hint: four decimal places are too many!)

6.6.4 Summary Data

The plans must also contain a table with a summary of pertinent UAS data such as dimensions, empty weight, motor/engine make and model.

6.6.5 Weight and Balance Data

The plans must also contain a weight and balance table with a summary of pertinent UAS equipment (motor, battery, payload, etc.), location from datum in metric units, moment arms and resultant moment of CG.

- All UAS must have a designated UAS datum indicated on the 2D drawings.
- All UAS drawings must indicate the following static CG margins: forward CG limit, aft CG limit and empty weight CG. Hint: Weight and Balance worksheet should correspond with static margins on 2D drawings.

6.6.6 Other Required Markings

The plans must be marked with the team's name and university or institute name.

6.7 SUBMISSION DEADLINES

The Design Report and 2D drawing plans must be electronically submitted to SAEINDIA no later than the date indicated on the Action. Neither the Organizer nor the SAEINDIA is responsible for any lost or misdirected reports, plans, or Server routing delays. SAEINDIA will not receive any paper copies of the reports through regular mail or email.

7.0. PRESENTATION GUIDELINES FOR PHASE 1

7.1 INTRODUCTION

Creating slides for presentation is a skill that is different from design report. PowerPoint Presentations skill is one of the effective visual communication tools that create the best first impression among the targeted audience. The SAEINDIA AEROTHON – ROTORCRAFT SYSTEMS CHALLENGE (UAS) – DESIGN, BUILD AND FLY CONTEST 2026 provides an excellent opportunity for students to master their presentation skills and showcase their project to Jury.

7.2 GENERAL

Presentation slides should effectively capture the work of the team. Follow a logical sound structure to organize the presentation. Here are some tips for making an effective presentation

- Plan and prepare your presentation professionally to deliver an effective message.

- Use visual points effectively, do not overwhelm your audience. A good PowerPoint presentation visual shouldn't complicate your message.
- Practice to perfection; rehearse your timing and delivery so that your points land as practiced with the Jury.
- Present with a relaxed calm and confident outward projection. Give your audience warmth, excitement, and energy.
- Avoid typos, cheesy clip art, and miscues like reading directly from your slides.

The team can identify preferably one or two team members to present their work in a compelling and influential manner to the Jury.

7.3 ORGANIZATION OF CONTENTS

Similar to the design report, the presentation must all contain the following,

- Explain the team's thought processes and engineering philosophy that drove them to their conclusions
- Detail the methods, procedures, and where applicable, the calculations used to arrive at the presented solution
- Covering these topics
    - UAS configuration selection
    - UAS design includes rotor arm, hub, landing gear, etc.
    - Subsystem Selection
    - UAS Performance
    - UAS C.G., stability and control
    - Computational Analysis
    - Other as appropriate

Note: The teams/students shall have all the CAD and CAE files in the PC or Laptop they will be using during the presentation. During the presentation, the teams can open the CAD model files and Computational analysis files in the appropriate software and present them to the jury for validation. The teams are expected to have the following documents ready during their presentation

1. **CAD files of the UAS**
2. **FEA input file along with format details**
3. **CFD input file along with format details**.

7.4 TIME LIMIT

While there is no limit on the number of PowerPoint slides, Teams needs to complete their presentation within the allotted 15 minutes. In case teams are unable to complete their whole presentation, they would be stopped at whatever point they are at after the end of 15 minutes. Post completion of the presentation, there will be 10 minutes of Q&A with Jury Panel.

8.0. REFERENCE BOOKS

1. Introduction to UAS Systems - Paul Fahlstrom and Thomas Gleason
2. Unmanned Aircraft Systems: UASS Design, Development and Deployment - Reg Austin
3. Advanced Aircraft Design: Conceptual Design, Analysis and Optimization of Subsonic Civil Airplanes - Egbert Torenbeek
4. Aircraft design: A conceptual approach - Daniel P. Raymer
5. Introduction to Flight- John D. Anderson
6. Fundamentals of Aerodynamics - John D. Anderson
7. Airplane Performance and Design - John D. Anderson
8. Flight stability and automatic control, Robert C. Nelson
9. Airframe stress analysis and sizing – Michael Chun-Yung Niu
10. Aircraft Structures, T.H.G. MEGSON (4th Edition)
11. https://docs.px4.io/master/en/concept/ (Ty Audronis , Designing Purpose-Built Drones for Ardupilot Pixhawk 2.1: Build drones with Ardupilot)

APPENDIX A

STATEMENT OF COMPLIANCE

Certification of Qualification

Team Name: ________________

University/Institute: ___________

Faculty Advisor: ______________

Faculty Advisor’s Email: _______

Statement of Compliance:

As Faculty Advisor, I certify that the registered team members are enrolled in collegiate courses. This team has designed the UAS for the SAE AEROTHON 2026 contest, without direct assistance from professional engineers, R/C model experts or pilots, or related professionals.

______________________________ ______________________________

Signature of Faculty Advisor Date

Team Captain Information:

Team Captain’s Name: ______________________________

Team Captain's E-mail: ______________________________

Team Captain’s Phone: ______________________________

<u>Note:</u>

A copy of this statement needs to be included in your Design Report as page 2

APPENDIX B

Engineering Change Request Form

Change Request	
Team Name:	Team ID:
Institute:	
Change Requester:	Date:
Change Requests information (Fill in appropriate information)	
Change Description:	
Details of Change:	
Alternates considered before selecting this change:	
Impact to previous Design:	
Why proposed change request should be approved? Explain	
What are the consequences if proposed change (s) is not implemented? Explain	

I have reviewed the information contained in this change request form and agree

Signature of Team Lead ***Signature of Faculty Advisor***

**Use additional sheets if the information cannot be accommodated in above form*

APPENDIX C

Geofence Information

The specific geofence parameters for Phase 2 will be made available to all qualified teams following the completion of Phase 1. These details will be shared in advance to ensure teams have sufficient time to prepare for the subsequent phase of the competition. Only teams that successfully qualify for Phase 2 will receive the necessary geofence information to proceed.